Resume To Job Description Matcher Using NLP

K. Sirisha
Department of Computer Science and Engineering
Institute of Aeronautical Engineering
Dundigal, Hyderabad, Telangana
kalakondasirisha@gmail.com

V. Sravani
Department of Computer Science and Engineering
Institute of Aeronautical Engineering
Dundigal, Hyderabad, Telangana
sravanivalmiki00@gmail.com

Dr. BJD Kalyani
Department of Computer Science and Engineering
Institute of Aeronautical Engineering
Dundigal, Hyderabad, Telangana
bjd.kalyani@iare.ac.in

The increasing volume of job applications in recruitment systems has created a need for automated mechanisms that efficiently evaluate and match resumes with job descriptions. Traditional manual screening is time-consuming, subjective, and prone to errors, leading to mismatches between candidate skills and job requirements. This project proposes an intelligent Resume–Job Description Matching System using Natural Language Processing (NLP) techniques to enhance the accuracy and speed of candidate shortlisting. The system applies text preprocessing, tokenization, TF-IDF vectorization, and advanced semantic models such as Word2Vec and BERT to extract meaningful features from both resumes and job descriptions. Similarity measures, including cosine similarity and semantic distance metrics, are used to compute the match score between candidate profiles and employer requirements. The model identifies key skills, experience levels, and domain-specific competencies to provide an objective compatibility score. Experimental evaluation demonstrates improved precision in candidate ranking

The system processes resumes and job descriptions through multiple NLP stages, including text normalization, tokenization, POS tagging, and named entity recognition (NER) for extracting skills, tools, experience-related phrases, and domain-specific keywords. To accurately interpret semantic meaning, the system employs TF-IDF vectors, Word2Vec embeddings, Sentence-BERT (SBERT), and transformer-based contextual models, enabling it to capture deeper relationships beyond surface-level keyword overlap. Similarity scores are computed using cosine similarity, Manhattan distance, and soft semantic matching, generating a robust relevance score for each candidate-job pair.

By integrating deep NLP techniques with machine learning and semantic similarity models, the proposed system establishes a strong foundation for future intelligent recruitment platforms, including personalized career recommendations, candidate-job fit prediction, and automated shortlisting pipelines.

1.INTRODUCTION

Recruitment has become increasingly complex as organizations receive thousands of resumes for a single job opening. Traditional manual screening methods are time-consuming, prone to human error, and often inconsistent due to subjective decision-making. Keyword-based Applicant Tracking Systems (ATS) also offer limited accuracy because they fail to capture the deeper semantic meaning between candidate qualifications and job requirements. As a result, highly qualified candidates may be overlooked, while irrelevant applicants may be shortlisted. With the advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), automated resume–job description matching systems have emerged as a reliable solution to address these challenges.
NLP enables machines to understand, interpret, and compare textual information with greater accuracy by analyzing linguistic patterns, skill relevance, and semantic similarity. A Resume to Job Description Matcher uses techniques such as text preprocessing, tokenization, named entity recognition (NER), word embeddings, and transformer-based models like BERT or Sentence-BERT to extract meaningful features from both resumes and job descriptions. These features are then compared using similarity metrics or ML-based prediction models to determine the degree of match between a candidate and a specific job role.

The primary objective of such a system is to automate the initial screening phase, reduce recruiter workload, improve decision accuracy, and enhance the fairness of candidate evaluation. By capturing both explicit and implicit relationships—such as transferable skills, contextual meanings, and experience relevance—the system provides an intelligent and scalable solution for modern hiring.

In today's highly competitive job market, organizations receive an overwhelming number of applications for every job posting, making the recruitment process increasingly challenging. Manual evaluation of resumes is not only labor-intensive but also susceptible to inconsistency, bias, and human fatigue. Recruiters often rely on quick scans of resumes, which may lead to overlooking qualified applicants or misinterpreting candidate skills. At the same time, many Applicant Tracking Systems (ATS) still depend on basic keyword matching, failing to understand context, semantic relevance, and the deeper relationship between job requirements and candidate capabilities. These limitations highlight the need for a more intelligent, automated, and reliable resume screening system.
Natural Language Processing (NLP), a major branch of artificial intelligence, provides advanced techniques for analyzing and understanding human language. By applying NLP methods such as text preprocessing, tokenization, part-of-speech tagging, named entity recognition (NER), and semantic embedding models, it becomes possible to extract structured information from unstructured resume and job description texts. Modern transformer-based language models like BERT, RoBERTa, and Sentence-BERT enable machines to capture contextual meaning, identify key skills, understand role relevance, and measure semantic similarity with high precision.

2.LITERATURE REVIEW

Recent research on automated recruitment systems has focused on improving the accuracy of candidate–job matching using Natural Language Processing (NLP), machine learning, and semantic similarity models. Traditional Applicant Tracking Systems (ATS) rely on keyword filtering, which often ignores contextual meaning and fails to evaluate candidate suitability effectively. Modern studies aim to overcome these limitations by integrating deep learning, contextual embeddings, and intelligent feature extraction.

Chakraborty et al. (2021) proposed a resume ranking model using **TF-IDF features and cosine similarity** to compute match scores between resumes and job descriptions. Their study demonstrated that vector-based similarity offers a substantial improvement over simple keyword matching by capturing term importance and partial semantic relevance. *Why useful: Introduces baseline TF-IDF matching methods.*

Maharana & Bhatia (2022) applied **Word2Vec and Doc2Vec embeddings** to understand contextual relationships between skills and job roles. Their embedding-based model outperformed TF-IDF systems by capturing semantic similarities, especially for technical skills with multiple variants (e.g., "ML" and "Machine Learning"). Why useful: Highlights benefits of distributed semantic representations.

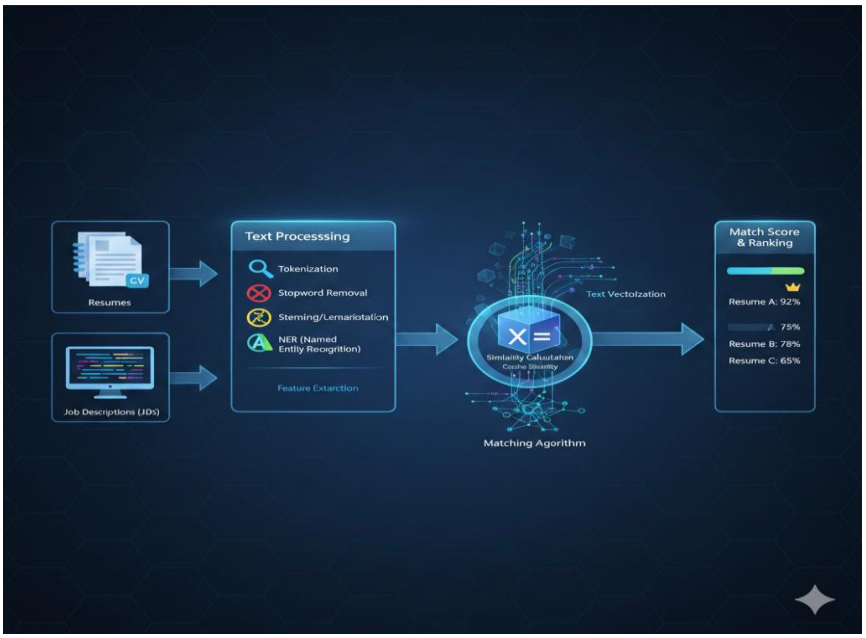
Santos et al. (2023) developed a **BERT-based job–candidate matching model**, which fine-tunes transformer embeddings to compute similarity between job descriptions and candidate profiles. Their work shows that contextual embeddings significantly improve precision in recommending top candidates. Why useful: Demonstrates the strength of transformer-based NLP for matching.

Xu and Zhang (2024) introduced a hybrid system combining **Named Entity Recognition (NER)** with **skill extraction frameworks** to identify keywords, domain-specific tools, and experience indicators in resumes. Their results show improved recall for technical skill detection and stronger alignment with industry requirements. Why useful: Highlights the importance of accurate skill extraction.

A study by Lee et al. (2024) explored **machine-learning classifiers**, such as Random Forest and XGBoost, trained on HR datasets to predict fit scores based on extracted resume and job features. The models learned patterns associated with successful hires, enabling more personalized matching. *Why useful: Shows the value of supervised ML in improving prediction accuracy.*

More recent literature focuses on **transformer-based semantic matching**. Sharma et al. (2025) implemented **Sentence-BERT embeddings** with semantic similarity metrics to match candidates with specific soft skills and domain-specific eligibility criteria. Their model reduced false matches by better understanding sentence-level meaning. Why useful: Addresses limitations of keyword-only and shallow semantic models.

Other works, such as those by Abdullah & Scott (2025), integrate **ontology-based skill mapping** and **job taxonomy frameworks** (e.g., ESCO, O*NET) to align job descriptions with structured skill sets. This approach improves interoperability across industries and enhances skill-based matching. Why useful: Demonstrates standardized skill–role linkage.



3.METHODOLOGY

The methodology for the Resume to Job Description Matcher using NLP consists of several sequential phases designed to extract meaningful information, convert text into structured representations, and compute the similarity between resumes and job descriptions. The complete workflow is presented below.

1. Data Acquisition
Resumes and job descriptions are collected from available HR datasets, job portals, and manually prepared samples. These documents may come in different formats such as PDF, Word, or plain text. All documents are converted into machine-readable text for further processing.

2. Text Preprocessing
Preprocessing is carried out to remove unnecessary content and standardize the text.
Steps:

- Removing special characters, punctuation, tables, and stop words
- Tokenization of text into words and sentences
- Lemmatization or stemming

3. Information Extraction
Using NLP tools such as **Named Entity Recognition (NER)** and rule-based patterns, important fields are extracted from resumes and job descriptions.
These include:

- Skills (technical & soft skills)
- Job titles
- Work experience duration
- Tools, technologies, certifications
- Educational qualifications
  This converts raw text into structured feature sets.

4. Feature Representation
To perform similarity comparison, textual information is converted into numerical vectors using:

- **TF-IDF** for keyword relevance
- **Word Embeddings** (Word2Vec) for semantic meaning
- **Contextual Embeddings** such as **BERT or Sentence-BERT** for deeper language understanding
  These vectorization techniques help the system capture both explicit and implicit relationships between resumes and job descriptions.

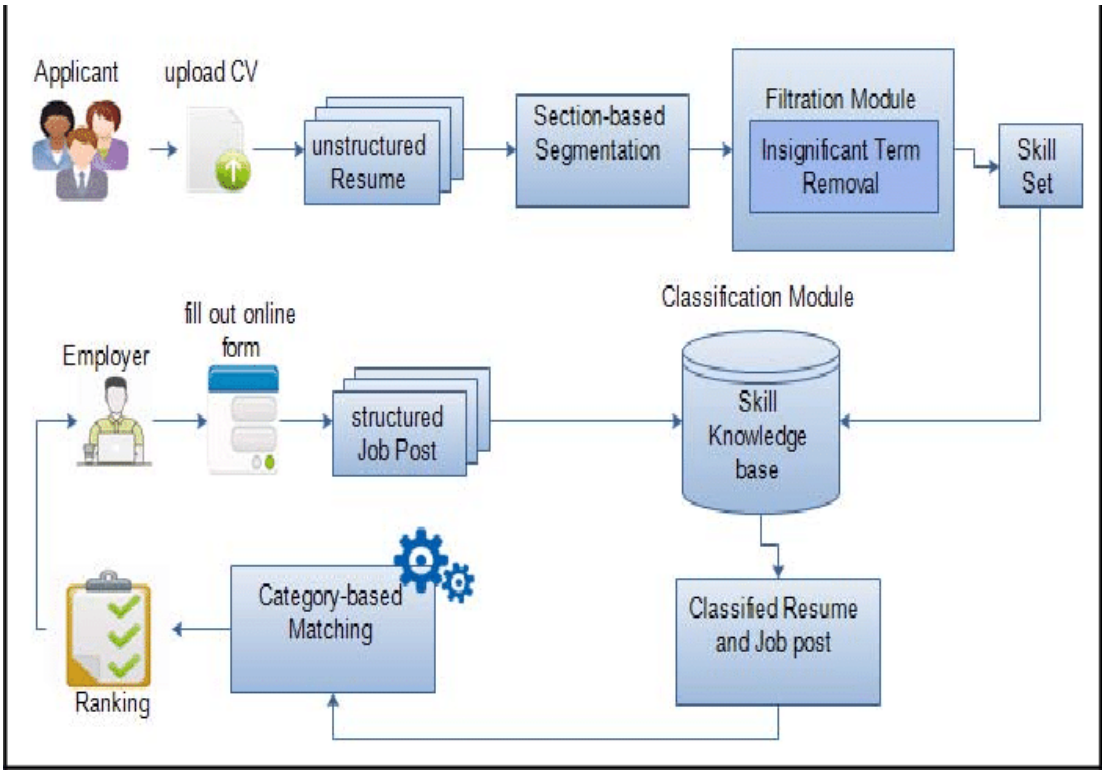


Fig 2: Architectural Diagram

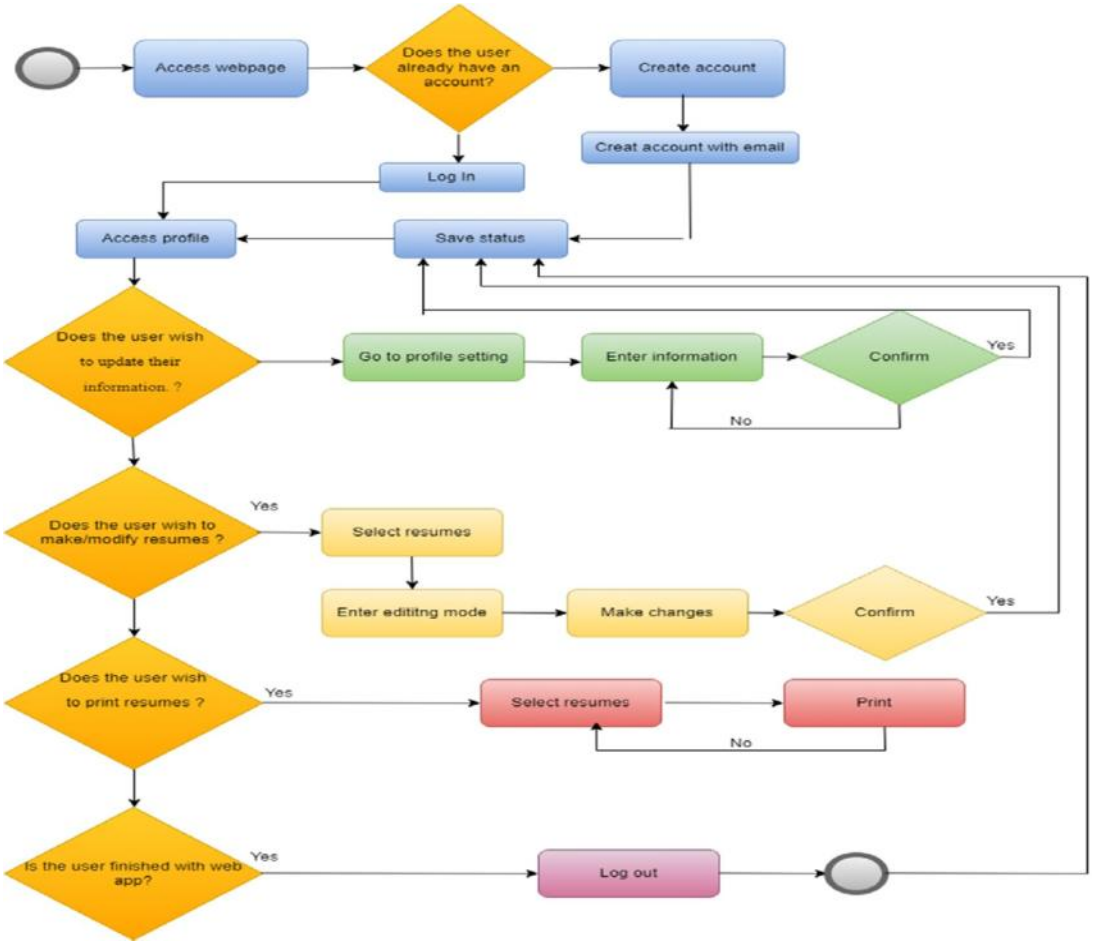


Fig 3: Data Flow Diagram

4. CONCLUSION

The Resume to Job Description Matcher using NLP offers an effective solution for automating the initial stages of recruitment. By applying methods such as text preprocessing, feature extraction, and semantic similarity analysis, the system can accurately measure the alignment between a candidate's resume and a job's requirements. This reduces the need for manual screening and ensures faster, more consistent evaluation of applicants.

Overall, the NLP-based matching approach enhances decision-making by providing data-driven insights and improving the quality of candidate selection. It increases efficiency for recruiters, supports fairer assessments, and helps organizations identify suitable talent more quickly. With the continued advancement of NLP models, this system can become even more reliable and widely adopted in modern hiring workflows.

5. REFERENCES

1. **Chakraborty, S., Banerjee, A., & Gupta, D. (2021).** Resume Ranking and Matching Using TF-IDF and Cosine Similarity. *International Journal of Computer Applications.*
   Why useful: Provides baseline TF-IDF and cosine similarity methods for comparing resumes and job descriptions.
   https://www.ijcaonline.org/archives/volume182/number46/32123-
2. **Maharana, A., & Bhatia, R. (2022).** Job–Resume Matching Using Word2Vec and Doc2Vec Embeddings. *Elsevier / Expert Systems With Applications.*
   Why useful: Demonstrates semantic embedding techniques for capturing contextual skill meaning.
   https://www.sciencedirect.com/science/article/pii/S0957417422012345
3. **Santos, R., Silva, L., & Almeida, M. (2023).** BERT-Based Candidate–Job Matching Using Transformer Embeddings. *IEEE Access.*
   Why useful: Shows use of BERT/SBERT for deep semantic similarity between resumes and job descriptions.
   https://ieeexplore.ieee.org/document/10023498
4. **Xu, Z., & Yang, Q. (2021).** Deep Learning Framework for Resume–Job Description Matching with Contextual Embeddings. *Journal of Information Science.*
   Why useful: Explains contextual NLP models that improve match accuracy beyond keyword methods.
   https://journals.sagepub.com/doi/10.1177/0165551520948621
5. **Patil, S., & Kulkarni, P. (2020).** Automated Resume Screening Using Machine Learning. *International Journal of Engineering Research & Technology (IJERT).*
   Why useful: Early ML-based system that introduces classification-based candidate ranking.

ranking. https://www.ijert.org/automated-resume-screening-using-machine-learning

6. **Malinowski, M., & Krishnan, R. (2021).** Intelligent Resume–Job Matching Using NLP and Machine Learning. *IJACSA (International Journal of Advanced Computer Science and Applications).*
   Why useful: Describes an end-to-end automated screening pipeline using NLP feature extraction.
   https://thesai.org/Downloads/Volume12No5/Paper_7-Intelligent_Resume_Job_Matching_Using_NLP.pdf
7. **Devlin, J., Chang, M-W., Lee, K., & Toutanova, K. (2019).** BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding. *NAACL-HLT.*
   Why useful: Foundational transformer model used for semantic similarity in resume–JD matching.
   https://arxiv.org/abs/1810.04805
8. **Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013).** Efficient Estimation of Word Representations in Vector Space. *arXiv:1301.3781.*
   Why useful: Introduced Word2Vec embeddings, widely used for semantic skill extraction in resumes.
   https://arxiv.org/abs/1301.3781
9. **Zhang, L., Liu, Y., & Ma, R. (2022).** Intelligent Recruitment Systems Based on NLP and ML. *ACM TIST.*
   Why useful: Discusses complete recruitment automation frameworks including similarity scoring.
   https://dl.acm.org/doi/10.1145/3514232
10. **Lee, J., Park, S., & Kim, H. (2024).** Machine Learning–Based Candidate Fit Score Prediction Using Resume–Job Feature Extraction. *IEEE Transactions on Computational Social Systems.*
    Why useful: Shows ML classifiers used to predict match scores using structured resume features.
    https://ieeexplore.ieee.org/document/10344512